

Surround-Screen-Based Display Systems in Immersive Visualization Laboratories: Applications, Benefits, and Lower-Cost Solutions

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ABSTRACT

Over the past 40 years, immersive visualization laboratories have existed in various forms across academia, industry, and government. Common to many of these laboratories has been the presence of surround-screen-based immersive display systems, such as CAVEs, powerwalls, tiled display systems, and related variations. With the rise of consumer off-the-shelf (COTS) virtual reality (VR) head-mounted displays (HMDs) over the past ten years, questions have arisen regarding the costs versus benefits of continuing to build, operate, and maintain surround-screen-based immersive display systems. This document highlights the benefits of continuing to promote these systems, both through a survey of existent systems post-2020 and by specifically showcasing low-cost approaches undertaken during a recent upgrade of a six-sided CAVE system. It will discuss applications unique to surround-screen-based immersive display systems, the advantages of these systems over COTS HMDs, and future needs and directions for these technologies to remain beneficial to users of XR technology.

Index Terms: Immersive Visualization Laboratory, CAVE, VR

1 INTRODUCTION

Over the past 30 years, surround-screen-based immersive display systems—such as CAVEs, Powerwalls, and Tiled Display Systems—have been the cornerstone of immersive visualization laboratory spaces. Originally, the CAVE was designed to address latency and cyber-sickness issues prevalent in the first wave of VR technologies during the 1990s, offering an alternative display architecture to head-mounted displays (HMDs) [10]. In recent years, significant advancements in commercial off-the-shelf (COTS) HMD technology have transformed the VR landscape. Devices like the Meta Quest, HTC Vive, and Valve Index have largely resolved the latency issues that plagued early HMD systems. Additionally, the emergence of online XR app distribution platforms, such as the Meta App Store and SteamVR, has made it easier for users to access and install XR applications directly on these devices. These advancements, combined with the drastically reduced cost of HMD technology since the 1990s, have made HMDs more accessible and user-friendly than ever before. Given these technological strides, the role and relevance of surround-screen-based display systems in modern immersive environments come into question: how can they adapt and continue to provide value in the era of high-performance, cost-effective HMDs?

This work presents findings on the current use of surround-screen-based immersive display systems in immersive visualization laboratories. It highlights applications unique to these systems, outlines the advantages they offer over COTS HMDs, and explores

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Figure 1: Six-Sided CAVE System upgraded in 2024 from original 2009 design

new, cost-effective approaches for developing or upgrading such systems. These approaches are illustrated through a case study of the recent upgrade of a six-sided CAVE system at a university research lab (Figure 1).

2 SURROUND-SCREEN-BASED DISPLAY SYSTEMS

After the initial advent of the CAVE system in 1992, several variations and alternate versions surfaced over several years. At the heart of these systems, images were typically presented on screens surrounding the users either via projectors or LCD panels. Projectors often serve as the display source for CAVE systems, while LCD panels usually serve as the source of tiled display systems. CAVE systems typically have anywhere between two and six sides. Some variations of these exist such as the StarCAVE, which expanded the shape of the traditional CAVE from being a cube to allowing for various surfaces at different angles and orientations around the user to act as displays [13]. Later, the CAVE2 system tiled several 3DTVs in a semi-circle around multiple users and offered higher resolution, brightness, and contrast than projectors due to its use of LCD panel technology [14]. This system also spawned different variations, such as the DSCVR system, which sought to reduce size and cost of the original system, while still maintaining a similar number of pixels per visual angle [29]. Another variation of this is the Reality Deck, which acts as a full room of tiled displays; however, it sacrifices 3D capabilities in lieu of higher resolution [27]. Tiled display systems typically have suffered from off-axis

viewing artifacts, such as ghosting in 3D viewing modes. More recently, direct view micro LED technology has promised to offer further improved brightness and contrast even when viewing off-angle [30]. This has been shown to be true from recent work in the development of a LED based CAVE system [25]. Other types of surround-screen-based immersive display systems include dome-based approaches, which sometimes use fish eye lens based projection to display spherically warped images onto spherical surfaces, resulting in a perspective-correct image [2]. The Allosphere is an extreme example of an immersive dome display [19]. In addition portable immersive display systems have been developed such as the IQ Station [31].

Traditionally, it has been a challenge to survey numbers of surround-screen-based immersive display systems in existence. Such systems come and go across academia, industry and government, and information about these systems is not always readily available to the public. In surveying current numbers of surround-screen-based immersive display systems through web searches, the authors have found over 50 sites publicizing the use of such systems¹. Furthermore, in searching Google Scholar for “CAVE virtual reality”, the authors found over 50 peer-reviewed research articles since 2020. In addition, through web searches, the authors have found a total of eight different surround-screen display manufacturers actively present in 2024. Based on these results, it is clear that the creation and use of surround-screen-based displays is still actively occurring, with many of these cases existing at immersive visualization laboratories. Given the rise of COTS VR HMDs, one may wonder why this is the case? The following section will describe some benefits of these systems over COTS VR that may continue to drive their adoption and use.

3 BENEFITS OF SURROUND-SCREEN-BASED SYSTEMS

Despite the rise of COTS VR and its potential impact on surround-screen-based immersive display systems, many IVLs remain committed to these types of systems and believe they still bring benefits that COTS VR HMDs cannot deliver. Benefits of surround-screen-based immersive display systems include:

3.1 Visibility of Other People

A benefit of CAVE style systems is the ability to easily see other people within the space. This allows for collaborative experiences between users without the typically required networked setup with COTS VR [32]. Optionally, some surround-screen-based systems allow for multiple users to experience their own stereoscopic tracked viewpoint, through the use of high refresh rate (240-480 hz) projectors and custom 3D glasses [23]. Unfortunately cost of these projectors remains high, and thus isn't commonly adopted. With the rise of what COTS manufacturers have dubbed “mixed reality” visibility of other people while using a COTS HMD has become more feasible. Nonetheless, it is not without its own logistical challenges due to the software development requirements involved. This form of “mixed reality” is, in actuality, a re-emphasis on video-passthrough based augmented reality displays in scenarios with multiple users in a co-located physical environment. In contrast, surround-screen-based systems allow users to simply walk into the display space to be co-located with others, eliminating the need for complex software or hardware solutions to achieve collaborative presence.

3.2 Visibility of One's Own Body

Another distinct advantage of a surround-screen-based system is the ability to see one's own body while immersed in the environment, which is typically not possible with an HMD. Although recent COTS HMDs offer video passthrough functionality, their field

of view remains limited, and users must still contend with wearing a device that obscures their face. The ability to see one's body in a surround-screen system enhances vestibular and kinesthetic sensory experiences compared to using a COTS VR HMD [1].

3.3 Higher Resolution

A third advantage of CAVE and tiled display systems are the typical higher resolution associated with these systems vs COTS VR. In particular, tiled display systems can reach gigapixel resolution and be used to visualize high resolution imagery in detail [28]. COTS HMDs continue to increase in resolution, but have plateaued recently around 2K x 2K pixels per eye, with some specialty devices, such as Pimax HMDs, reaching higher resolutions.

3.4 Field of View

Due to their design, COTS HMDs typically have a field of view that is drastically less than the human visual system's field of view of 180 degrees [34]. Most COTS HMDs exist in the 90-120 degree range, thus falling short between 60-90 degrees. Since the end user is typically wearing a pair of 3D glasses, or no glasses at all, surround-screen-based display systems afford a closer match to the traditional human field of view.

3.5 Less Tethered & Increased Comfort

To experience 3D content, surround-screen-based systems that generate stereoscopic images require users to wear only some form of active or passive 3D glasses as opposed to a full HMD. This traditional glasses form factor weighs significantly less than a HMD, thereby reducing overall fatigue while engaging with the content. Furthermore, one could argue that the act of wearing a COTS VR HMD detracts from the VR experience and diminishes the sense of presence due to the distraction of having the device physically on one's head [33]. Additionally, research comparing low-cost CAVE systems to HMDs in a visual search task has shown that CAVE users report significantly fewer symptoms of cybersickness compared to HMD users [15]. Overall, surround-screen-based systems leave end users far less encumbered and more comfortable, both physically and mentally.

4 APPLICATIONS UNIQUE TO SURROUND-SCREEN-BASED SYSTEMS

Arguably, the rise of COTS VR has slightly shifted attention away from application development for surround-screen-based systems. However, simultaneously, new application areas for these systems have emerged, offering capabilities that cannot be as effectively achieved using COTS HMDs. Given the differences outlined above between the two types of systems, it is valuable to consider the types of VR applications best suited for each. First-person VR experiences naturally align well with COTS HMDs, as these devices are specifically designed to provide a first-person viewpoint by placing displays directly in front of a user's eyes. While surround-screen-based systems can also deliver a first-person perspective, they may be better suited to applications that benefit from a third-person perspective, such as visualizing complex 3D environments or structures—like a molecular model [20]. Examples of applications better suited to surround-screen-based systems include:

4.1 Virtual Production

One application that benefits from surround-screen-based displays has been coined “virtual production” [17] [16]. One aspect of virtual production is a film technique whereby the background of a filmed scene is displayed using a surround-screen-based display. This effectively allows for any background to exist behind a filmed set of actors. This technique, recently used by the show, “The Mandalorian”, has seen increased use cases in recent years within display spaces such as CAVEs and tiled display walls.

¹<https://www.thecaav.org/cave-list>



Figure 2: Outreach experiences within surround-screen-based displays remain logistically easier than with COTS HMDs

4.2 Immersive 3D Collaborative Visualization

Collaborative visualization involves analyzing a dataset together with other people, and researchers have shown this can benefit from occurring in an immersive display context [3], [4]. Collaborative visualization can be achieved with COTS HMDs using mixed reality, or networked VR approaches, but these applications require a more complex software framework through the networking requirements to achieve the collaborative shared virtual environment. Furthermore, they have the challenge and requirement of transmitting some form of avatar representation over the networked setup if users are not colocated in the same space. Several collaborative 3D immersive visualization software tools have been developed that provide a starting point for adopting new surround-screen-based systems for these applications [35], [9].

4.3 Education and Outreach

Several researchers continue to use surround-screen-based systems for education [12] [37] [11] [18] [24]. While HMDs have also undoubtedly seen a rise in educational applications, surround-screen-based systems offer distinct advantages, particularly in group settings. For one, surround-screen systems provide a more accessible entry point for groups of students—typically requiring only a pair of 3D glasses rather than the individualized fitting process necessary for HMDs (Figure 2). Additionally, certain VR platforms, such as Meta’s App Lab, restrict the simultaneous use of a single app instance per account. This means that enabling multiple students to use the same app requires separate accounts, which can pose significant logistical challenges. Furthermore, during outreach events, it can be advantageous for observers to view participants’ interactions with virtual content. While COTS HMDs achieve this by casting the first-person view to a display, doing so for multiple HMDs necessitates multiple additional displays. Configuring and ensuring smooth operation of these casts for simultaneous use introduces further logistical complexity to such events.

4.4 HMD + Surround Screen Display Systems

Another possibility that immersive surround display systems provide is the ability to combine its display with a COTS HMD. This combination can be used for applications such as design review, whereby an audience could don 3D glasses and view the CAVE that shows an over the shoulder view of the person in the HMD. It is likely that many of the possible application areas remain yet to be discovered when combining these technologies together.

5 MODERN LOW-COST APPROACHES: A SIX-SIDED CAVE UPGRADE

This section will discuss cost-reducing approaches for surround-screen-based systems by looking specifically at a recently conducted upgrade to the author’s six-sided CAVE system. The system measures nine and a half feet cubed, is rear projected on all slides, and uses a sliding door that allows entrance and exit from the system, along with projection onto the sixth wall. Originally, the system was equipped with twelve Digital Projection Titan 1080P 3D projectors, two projectors per nine and a half foot wall, with each wall having a display resolution of 1920x1920 pixels in 3D active stereo at 60 frames per second. The projection display system could optionally allow for two simultaneously tracked users from two distinctly tracked positions at 30 frames per second. A six degree of freedom tracking system operates via an Intersense IS900 Ultrasonic tracking system. Miniature ultrasonic emitters are discretely mounted around the corners of the CAVE walls, with paired microphones for those emitters existing on the associated IS900 wand and head-mount tracking hardware. The floor and ceiling projectors bounce images off of symmetrically oriented mirrors that reflect the associated images onto the floor and ceiling screens. Originally, active 3D stereo was accomplished via RealD 3D active shutter stereo glasses that retailed at \$175 each and Stereographics EXXR infrared 3D emitters retailing at \$830 each, with six emitters mounted around the outside of the CAVE walls on the projector towers. The 3D emitters and glasses originally totaled around \$6,700. The system was originally driven by a \$70,000 cluster of seven PCs each with two Quadro 5000 GPUs.

An upgrade was recently completed that replaced the PC cluster, projectors, and 3D glasses and emitters. The PC cluster was swapped to seven new systems running Windows 11 with new GPUs (nVidia Quadro A4000s with Quadro Sync II to allow for a synchronized 3D signal across the GPU cluster). Each system has 64 GB of RAM, AMD Ryzen 7 G-Series CPUs and two Samsung SSDs. The cost of the PC cluster upgrade was about \$25,000. At the same time that COTS VR has become mainstream with lower cost HMDs, projector technology has also seen vast improvements and reductions in cost. Some of the most recent advances in projector technologies have been driven by the increase in popularity of golf simulators [5]. During the CAVE upgrade, the prior projectors were replaced with BenQ LK936ST short throw projectors (Figure 3). These projectors contain 3D sync-in sync-out VESA ports that allow for 3D synchronization across the set of projectors. The projectors remain the same stereo 1920x1920 pixel resolution as the original Digital Projection projectors; however, the cost for the BenQ projectors versus the original cost for the Digital Projection projectors is reduced by about a factor of 16 (\$75,000 per Digital Projection projector vs. \$4,800 per BenQ projector), for a projector cost difference of \$842,400. In addition, the BenQ projectors can operate at 120 frames per second, which positions the display system with a refresh rate equivalent to the maximum refresh rates of the majority of COTS HMD systems. All six prior 3D IR emitters were replaced with one Volfon Edge RF emitter. Volfon Edge RF 3D active shutter glasses were purchased to replace the prior 3D glasses, with the new emitter and glasses totaling about \$1200. The emitter is connected to the VESA 3D sync out port on one projector. All 12 projectors have a VESA cable connected to the 3D sync-in port. The VESA cables are connected to VESA to BNC converters and the BNC cables connect to a BNC splitter, which takes as input a single BNC cable connected to the Quadro Sync II card of “server” node in the cluster. Table 1 summarizes the cost reduction from the original system.

6 DISCUSSION

The cost of the system upgrade totaled approximately \$84,000. This is a drastic (factor of 10x) reduction in cost from the origi-



Figure 3: BenQ LK936ST 3D Projector

	2009 System	2024 System	Cost Difference
Projectors	\$900,000	\$57,600	-\$842,400
Servers	\$70,000	\$25,109	-\$44,891
3D Glasses	\$6,730	\$1,231	-\$5,499
Total	\$976,980	\$83,940	-\$892,590

Table 1: Cost Comparison Table for 6-Sided CAVE Upgrade - 12 projectors, 7 servers, and 10 pairs of 3D glasses with 1 new emitter we purchased for the upgrade

nal system projection, cluster, and emitter costs of \$976,980. While the footprint of the original system was reused, along with some cabling, and the IS900 tracking system, the total cost of the upgrade is likely not far off from the cost to replicate the system, given available space. The IS900 Tracking System originally cost \$27,000; but with recent lower cost 3D tracking systems such as Vive Light-houses, the cost of 3D tracking could be much lower for a surround-screen-based display system. Some systems have even successfully used Microsoft Kinect type devices for reliable 3D tracking [29].

Despite the ability to upgrade a CAVE system at a fraction of the original cost, there remains some challenges for new surround-screen-based system adoption. The BenQ LK936ST model projector is one of only a very small number of affordable projectors that have the flexible 3D sync in/out ports to allow for VESA based 3D stereo. There are potentially newer forms of 3D stereo transmission, such as directly over display port, but many low cost projectors are not equipped with display port inputs. Another current downfall is that while costs have reduced for projectors that display an (arguably) previous generation 1920x1920 (i.e. HD) resolution, costs for stereoscopic 4K projectors remains high, and only available via specialty manufacturers such as Digital Projection, Christie, and Barco. Thus, there remains a need for lower cost 4K projectors that support 3D stereoscopy. When considering tiled based systems, unfortunately 3D TV availability for CAVE2 style systems left the consumer market in 2015, thus requiring expensive, specialized manufacture from companies such as Planar Systems.

Prior work has contributed to the effort of comparing surround-screen-based displays to COTS HMD systems for a variety of tasks [6] [26] [7]. One work has shown that if successfully establishing an equivalent framework between a CAVE2 and multiple networked HMDs, that when performing network analysis tasks, no

significant differences were found in accuracy and the networked HMDs actually resulted in faster task analysis [8]. As this was a specific comparison between a CAVE2 style system and HMDs for a specific network analysis task, there remains a lot of additional research needed to compare and assess general differences between HMD and surround-screen-based displays. Now that the authors have the ability to match refresh rates between their CAVE system and current COTS HMDs, and due to the fact that the CAVE is 6-sided so therefore has no differences in field of regard from a HMD, the authors plan to conduct further research comparing CAVEs to COTS HMDs for different immersive visualization tasks.

A final challenge with surround-screen-based display systems is adopting software solutions to these systems. Unity has become a de-facto standard for 3D virtual environment creation for COTS HMDs, and there exists solutions for adopting Unity to surround-screen-based systems; however, it remains a challenge to keep these solutions up-to-date with changes made to the Unity runtime environment [36], [21]. In addition, some of these solutions have associated costs. The Unreal Engine offers built in support for multiple displays through its nDisplay framework, and may serve as a more robust solution for driving applications for these systems.

When reconsidering the initial question of how surround-screen-based display systems remain popular, besides the aforementioned applications and benefits, the general rise of COTS VR may have afforded a new generation of people discovering CAVEs and tiled display systems for the first time. This would, in particular, explain the increase in surround-screen-based manufacturers at present day and the large number of systems that still exist. There have also been glimpses from some of the big tech companies work in this area, for example Google's Project Starline, which could further drive general expansion [22].

7 CONCLUSION

Immersive visualization laboratories (IVLs) have faced a reckoning with the rise of COTS VR devices over the past ten years. That being said, and as outlined in this document, these spaces provide unique opportunities and enhanced possibilities for federal funding, outreach, and technology development beyond what can be achieved with COTS HMDs alone. Furthermore, maintaining both types of systems continues to create benefits for IVLs, offering the unique ability to perform comparative studies between surround-screen-based displays and COTS HMDs. The continued reduction in the cost of display technologies, such as projectors, may improve the feasibility for researchers to re-adopt immersive VR systems like CAVEs. How long surround-screen-based displays will remain relevant is an open question, especially as COTS HMDs continue to shrink in size and improve in performance. However, surround-screen-based display systems currently allow for the study of what might eventually become the idealized HMD form factor: a lightweight, glasses-like device with a wide field of view. Until COTS HMD technology reaches that point, and for the many reasons outlined in this document, surround-screen-based display systems remain invaluable in IVLs.

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